WORKED SOLUTION

# STEP II 1989, Question 15

Probability and Statistics • Geometric Probability

**STEP II 1989, Question 15**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q15

## The key idea

A point chosen "at random on the perimeter" is uniform with respect to length, so everything starts from the total length $\pi + 2$. The triangle always has a vertex at the centre $O$, so its area is $\frac{1}{2}ab\sin C$, where $a$ and $b$ are the distances from $O$ and $C$ is the angle at $O$. Each conditional case then reduces to a simple expectation, and the law of total expectation stitches them together.

## Solution

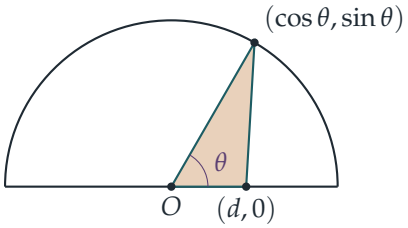
Put $O$ at the origin, with the diameter along the $x$-axis from $(-1, 0)$ to $(1, 0)$ and the arc in $y \geq 0$. The perimeter has length $2 + \pi$: the diameter contributes 2 and the arc $\pi$. Since each point is uniform along the perimeter,

$$\mathrm{P}(\text{a point lies on the diameter}) = \frac{2}{\pi + 2}, \qquad \mathrm{P}(\text{a point lies on the arc}) = \frac{\pi}{\pi + 2}.$$

Given that it lies on the arc, the point is $(\cos\theta, \sin\theta)$ with $\theta$ uniform on $[0, \pi]$. Given that it lies on the diameter, it is $(d, 0)$ with $d$ uniform on $[-1, 1]$.

The two points are independent, and "exactly one on the diameter" can happen with either the first or the second point on the diameter, so

$$\mathrm{P}(\text{exactly one on the diameter}) = 2 \cdot \frac{2}{\pi + 2} \cdot \frac{\pi}{\pi + 2} = \frac{4\pi}{(\pi + 2)^2}.$$



**Part (i)**

Let one point be $(d, 0)$ on the diameter and the other $(\cos\theta, \sin\theta)$ on the arc. The sides from $O$ have lengths $|d|$ and 1, and the angle between them is $\theta$ (if $d < 0$ the angle is $\pi - \theta$, which has the same sine). So

$$A = \tfrac{1}{2}\,|d| \sin\theta.$$

Given this case, $d$ and $\theta$ are independent, so the expectation of the product is the product of the expectations. Since $d$ is uniform on $[-1, 1]$, $|d|$ is uniform on $[0, 1]$ and $\mathrm{E}|d| = \frac{1}{2}$. Also

$$\mathrm{E}(\sin\theta) = \frac{1}{\pi}\int_0^\pi \sin\theta\,\mathrm{d}\theta = \frac{2}{\pi}.$$

Therefore

$$\mathrm{E}(A \mid \text{exactly one on the diameter}) = \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{2\pi},$$

as required.

**Part (ii)**

Now both points are on the arc, at angles $\theta_1$ and $\theta_2$, independent and uniform on $[0, \pi]$. Both are at distance 1 from $O$, and the angle between them at $O$ is $X = |\theta_1 - \theta_2|$, which lies in $[0, \pi]$. So

$$A = \tfrac{1}{2}\sin X.$$

? **Ask yourself** Why can we use the given density for $X$?

The angle $\theta$ measures arc length along the unit semicircle, so choosing two points uniformly on the arc is the same as choosing two points uniformly on a line segment of length $\pi$. The angle $X$ is exactly the distance between them, so $X$ has density $2(\pi - x)/\pi^2$ on $[0, \pi]$. Hence

$$\mathrm{E}(\sin X) = \frac{2}{\pi^2}\int_0^\pi (\pi - x)\sin x\,\mathrm{d}x.$$

Splitting the integral, $\int_0^\pi \pi \sin x\,\mathrm{d}x = 2\pi$, and integrating by parts with $u = x$,

$$\int_0^\pi x\sin x\,\mathrm{d}x = \Big[-x\cos x\Big]_0^\pi + \int_0^\pi \cos x\,\mathrm{d}x = \pi + 0 = \pi.$$

So $\int_0^\pi (\pi - x)\sin x\,\mathrm{d}x = 2\pi - \pi = \pi$, giving $\mathrm{E}(\sin X) = \dfrac{2}{\pi}$ and

$$\mathrm{E}(A \mid \text{neither on the diameter}) = \frac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{\pi},$$

as required.

#### The expected value of $A$

There is a third case: if both points lie on the diameter, the "triangle" is degenerate (all three points are on the $x$-axis), so $A = 0$. The probability that neither point is on the diameter is $(\frac{\pi}{\pi+2})^2$. By the law of total expectation,

$$\begin{aligned} \mathrm{E}(A) &= \frac{4\pi}{(\pi+2)^2} \cdot \frac{1}{2\pi} + \frac{\pi^2}{(\pi+2)^2} \cdot \frac{1}{\pi} + \frac{4}{(\pi+2)^2} \cdot 0 \\ &= \frac{2+\pi}{(\pi+2)^2} \\ &= \frac{1}{2+\pi}, \end{aligned}$$

which is the result we wanted.

## Checks

- The three case probabilities $\frac{4}{(\pi+2)^2}$, $\frac{4\pi}{(\pi+2)^2}$ and $\frac{\pi^2}{(\pi+2)^2}$ add up to $\frac{(2+\pi)^2}{(\pi+2)^2} = 1$.
- $\mathrm{E}(A) \leq \frac{1}{2}$, the largest possible area (two radii at right angles), as it must be; in fact $\frac{1}{2+\pi} \approx 0.1945$.
- A Monte Carlo simulation of $2 \times 10^6$ pairs of points gave $\mathrm{P}(\text{exactly one}) \approx 0.4757$ (exact 0.4754), conditional means 0.1591 and 0.3184 (exact $\frac{1}{2\pi} \approx 0.1592$ and $\frac{1}{\pi} \approx 0.3183$), and $\mathrm{E}(A) \approx 0.1944$.

## Common slips

- Forgetting the factor 2 in $\mathrm{P}(\text{exactly one})$: either point can be the one on the diameter.
- Treating "on the diameter" as having probability $\frac{1}{2}$: the choice is uniform by length, not by which piece of the boundary.
- Using $\sin\theta_1 \sin\theta_2$ or similar in part (ii) instead of the sine of the angle *between* the radii.

## Beyond STEP

Splitting an expectation over a partition of cases, $\mathrm{E}(A) = \sum_i \mathrm{E}(A \mid B_i)\,\mathrm{P}(B_i)$, is the law of total expectation, one of the workhorses of first-year probability; its general form, $\mathrm{E}(A) = \mathrm{E}\big(\mathrm{E}(A \mid Y)\big)$, turns up everywhere from branching processes to Markov chains. The density of $X$ quoted in part (ii) is itself a nice exercise: it comes from the area of a strip in the square $[0, \pi]^2$. Questions like this belong to geometric probability; Grimmett and Welsh, *Probability: An Introduction*, covers conditional expectation and joint densities at the right level.

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